MODEL RESEARCH

Title: Model Research on Machine Learning Techniques for Cyber Threat Detection

Project Title: Development of Interactive Cyber Threat Visualization Dashboard

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Introduction

Cybersecurity plays a crucial role in protecting systems, networks, and data from cyber threats and attacks.

Machine learning models are widely used to detect, classify, and prevent cyber attacks by analyzing large volumes of security data.

This document presents a research study on various machine learning models commonly applied in cybersecurity, along with their accuracy, advantages, and limitations.

The models were researched using AI-based prompting techniques.

Model 1: Logistic Regression

Prompt Used: Explain Logistic Regression model, its use in cybersecurity, and its accuracy in detail explanation

Model Explanation (Neat & Simple)

Logistic Regression is a **supervised machine learning model** used for **classification problems**.
It predicts whether an event belongs to one category or another, such as **attack vs no attack**.

In cybersecurity, Logistic Regression is commonly used because it is **simple, fast, and easy to understand**.

Use of Logistic Regression in Cybersecurity

- Phishing email detection
- Spam classification
- Intrusion detection (normal vs malicious traffic)

Accuracy:

The accuracy of Logistic Regression is **generally moderate to high** when:

- The data is clean
- The problem is simple and linearly separable

In cybersecurity tasks like phishing detection, Logistic Regression can achieve **around 85%–90% accuracy** depending on the dataset.

Advantages

- Easy to implement and understand
- Fast training and prediction
- Works well for binary classification

Limitations

- Not suitable for complex patterns
- Performance drops with highly non-linear data

Model 2: Decision Tree

Prompt Used: Explain the Decision Tree model, its application in cybersecurity, and its accuracy

Model Explanation (Neat & Simple):

A Decision Tree is a machine learning model that works like a **flowchart**.
It makes decisions by asking a series of **yes/no questions** based on data features.

In cybersecurity, Decision Trees help in identifying whether an activity is **safe or malicious** by following decision rules.

Use of Decision Tree in Cybersecurity

- Malware detection
- Intrusion detection systems (IDS)
- Network traffic classification

Accuracy

Decision Trees usually provide **good accuracy** when trained on structured data.
In cybersecurity applications, Decision Trees can achieve **around 88%–92% accuracy**, depending on data quality.

However, accuracy may decrease if the model becomes too complex.

Advantages

- Easy to understand and visualize
- Handles both numerical and categorical data

- Requires minimal data preparation

Limitations

- Can overfit the data
- Less stable with small changes in data

Model 3: Random Forest

Prompt Used: Explain the Random Forest model, its use in cybersecurity, and its accuracy

Model Explanation (Neat & Simple)

Random Forest is an **ensemble machine learning model** made up of many decision trees. Instead of relying on a single tree, it combines results from multiple trees to make a more **accurate and reliable prediction**.

In cybersecurity, Random Forest is widely used because it can handle **complex attack patterns** effectively.

Use of Random Forest in Cybersecurity

- Intrusion detection
- Malware and ransomware detection
- Fraud and anomaly detection

Accuracy

Random Forest generally provides **high accuracy** in cybersecurity applications.
It can achieve **90%–95% accuracy** when trained on well-prepared datasets.

Its ensemble nature helps reduce overfitting and improves overall performance.

Advantages

- High prediction accuracy
- Reduces overfitting compared to Decision Trees
- Works well with large datasets

Limitations

- Slower compared to simpler models
- Less interpretable than single decision trees

Model 4: Support Vector Machine (SVM)

Prompt Used: Explain Support Vector Machine (SVM), its application in cybersecurity, and its accuracy

Model Explanation (Neat & Simple)

Support Vector Machine (SVM) is a machine learning model used for **classification and pattern recognition**.
It works by finding the **best boundary (hyperplane)** that separates data into different categories, such as **malicious vs normal traffic**.

SVM is effective in cybersecurity because it can handle **high-dimensional and complex data**.

Use of SVM in Cybersecurity

- Malware classification
- Network intrusion detection
- Spam and phishing detection

Accuracy

SVM models usually achieve **high accuracy** when properly tuned.
In cybersecurity applications, SVM can reach **88%–94% accuracy**, depending on the dataset and kernel used.

SVM performs especially well when the data has clear separation.

Advantages

- Effective in high-dimensional spaces
- Works well with complex data patterns
- Good generalization capability

Limitations

- Computationally expensive
- Requires careful parameter tuning

Model 5: Naive Bayes

Prompt Used: Explain the Naive Bayes model, its use in cybersecurity, and its accuracy

Model Explanation (Neat & Simple)

Naive Bayes is a **probability-based machine learning model** that applies Bayes' Theorem. It assumes that all features are **independent**, which makes the model simple and fast.

In cybersecurity, Naive Bayes is commonly used for **text-based attack detection** such as spam and phishing emails.

Use of Naive Bayes in Cybersecurity

- Spam filtering
- Phishing email detection
- Malware classification

Accuracy

Naive Bayes usually provides **good accuracy** for text-based cybersecurity tasks.
In phishing and spam detection, it can achieve **85%–92% accuracy** depending on the dataset.

Despite its simplicity, it performs well in many real-world applications.

Advantages

- Fast and efficient
- Works well with large datasets
- Easy to implement

Limitations

- Assumes feature independence
- Less accurate for complex relationships

Model 6: K-Means Clustering

Prompt Used: Explain K-Means clustering, its application in cybersecurity, and its accuracy in simple terms.

Model Explanation (Neat & Simple)

K-Means is an **unsupervised machine learning model** used to group similar data points into clusters.
Unlike classification models, it does not use labeled data. Instead, it finds patterns by grouping similar behaviors together.

In cybersecurity, K-Means helps in identifying **unusual or suspicious activities** by separating normal behavior from abnormal behavior.

Use of K-Means in Cybersecurity

- Anomaly detection
- Network traffic analysis
- User behavior analysis

Accuracy

Since K-Means is an **unsupervised model**, accuracy is measured differently compared to classification models.
In cybersecurity applications, K-Means can effectively identify abnormal patterns with an **approximate effectiveness of 80%–88%**, depending on how well the data is clustered.

Advantages

- Simple and easy to implement
- Works well for large datasets
- Useful for anomaly detection

Limitations

- Requires the number of clusters to be defined in advance
- Sensitive to initial cluster selection

Model 7: Isolation Forest

Prompt Used: Explain the Isolation Forest model, its use in cybersecurity, and its accuracy in simple terms.

Model Explanation (Neat & Simple)

Isolation Forest is a machine learning model designed specifically for **anomaly detection**. Instead of learning normal behavior, it works by **isolating unusual data points**, which makes it very effective for detecting rare cyber attacks.

In cybersecurity, it is useful because most attacks are **abnormal compared to normal system behavior**.

Use of Isolation Forest in Cybersecurity

- Intrusion detection
- Insider threat detection
- Fraud and anomaly detection

Accuracy

Isolation Forest provides **high effectiveness** in detecting anomalies.
In cybersecurity use cases, it can achieve **around 85%–93% detection accuracy**, especially when identifying rare or unknown attacks.

Advantages

- Excellent for anomaly detection
- Works well with large datasets
- Does not require labeled data

Limitations

- Less interpretable results
- Performance depends on parameter tuning

Model 8: Neural Networks

Prompt Used: Explain Neural Networks, their application in cybersecurity, and their accuracy

Model Explanation (Neat & Simple)

Neural Networks are machine learning models inspired by the **human brain**.
They consist of layers of connected nodes that learn complex patterns from data.

In cybersecurity, Neural Networks are used to detect **advanced and sophisticated cyber threats** that simpler models may miss.

Use of Neural Networks in Cybersecurity

- Malware detection
- Advanced intrusion detection
- Behavioral analysis

Accuracy

Neural Networks can achieve **very high accuracy** when trained on large datasets.
In cybersecurity applications, they often reach **92%–97% accuracy**, depending on data quality and model design.

Advantages

- Handles complex and non-linear data
- High detection accuracy
- Suitable for advanced threats

Limitations

- Requires large amounts of data

- Computationally expensive
- Harder to interpret

Model 9: Gradient Boosting

Prompt Used: Explain the Gradient Boosting model, its use in cybersecurity, and its accuracy

Model Explanation (Neat & Simple)

Gradient Boosting is an **ensemble machine learning model** that builds models sequentially. Each new model focuses on correcting the mistakes made by previous models, which improves overall accuracy.

In cybersecurity, Gradient Boosting is used to detect **complex and hidden attack patterns**.

Use of Gradient Boosting in Cybersecurity

- Cyber threat classification
- Malware detection
- Intrusion detection

Accuracy

Gradient Boosting models typically provide **very high accuracy**.
In cybersecurity applications, they can achieve **90%–96% accuracy**, especially with properly tuned parameters.

Advantages

- High predictive performance
- Handles complex data patterns
- Reduces bias

Limitations

- Computationally expensive
- Slower training time

Model 10: Anomaly Detection Models

Prompt Used: Explain anomaly detection models, their role in cybersecurity, and their accuracy.

Model Explanation (Neat & Simple)

Anomaly Detection models are designed to identify **unusual or unexpected behavior** in data.
Instead of detecting known attacks, these models focus on finding patterns that **deviate from normal behavior**.

In cybersecurity, anomaly detection is critical for identifying **zero-day attacks** and unknown threats.

Use of Anomaly Detection Models in Cybersecurity

- Zero-day attack detection
- Network intrusion detection
- Insider threat detection

Accuracy

Anomaly Detection models provide **good detection accuracy**, especially for unknown threats.
In cybersecurity systems, they typically achieve **85%–95% detection accuracy**, depending on the data and method used.

Advantages

- Detects previously unknown attacks
- Works without labeled data
- Essential for modern security systems

Limitations

- Can generate false positives
- Requires careful tuning

Comparative Summary of Cybersecurity Models

Model Name	Model Type	Cybersecurity Application	Typical Accuracy	Key Strength
Logistic Regression	Supervised	Phishing & intrusion detection	85–90%	Simple and fast
Decision Tree	Supervised	Malware & intrusion detection	88–92%	Easy to interpret
Random Forest	Supervised	Threat and fraud detection	90–95%	High accuracy
Support Vector Machine (SVM)	Supervised	Malware classification	88–94%	Handles complex data
Naive Bayes	Supervised	Spam & phishing detection	85–92%	Fast for text data

Model Name	Model Type	Cybersecurity Application	Typical Accuracy	Key Strength
K-Means Clustering	Unsupervised	Anomaly detection	80–88%	Pattern discovery
Isolation Forest	Unsupervised	Insider threat detection	85–93%	Anomaly-focused
Neural Networks	Deep Learning	Advanced threat detection	92–97%	Learns complex patterns
Gradient Boosting	Supervised	Cyber threat classification	90–96%	High performance
Anomaly Detection Models	Unsupervised	Zero-day attack detection	85–95%	Detects unknown threats

Conclusion:

Different machine learning models are applied in cybersecurity based on the type of threat and data availability.
Supervised models help detect known attacks, while unsupervised and anomaly detection models are useful for identifying unknown or zero-day threats.